Mississippi Valley State University

# CS 341 — Discrete Structures

Practice Set 26: Approved Solutions

*Existence Proofs*

Covers Chartrand & Zhang, Section 3.6

**Status of this set**

**Ungraded practice.** Solutions released with it. Attempt everything first.

# Part I: How to Practice This Material

**Two jobs, very different costs**

**Prove** $\exists x \in S,\ R(x)$**:** name one element and verify it.
**Disprove** $\exists x \in S,\ R(x)$**:** argue that $R(x)$ fails for *every* $x \in S$.
The first is often a one-liner. The second is a genuine universal argument — testing values does not do it.

**Three mistakes to expect**

1. **Naming a witness without verifying it.**
2. **"Disproving" an existence claim by testing a few values.**
3. **Choosing a witness too late.** In $\exists r\ \forall s$, the $r$ must work for every $s$ at once.

# Part II: Practice Problems

## Section A — Constructive existence

**1.** Prove each by exhibiting a witness and verifying it.

(a) There exists an integer $n$ such that $50 - n^2 < 0$.

(b) There exists a real number $x$ such that $x^2 - 3x - 10 = 0$.

(c) There exist integers $m$ and $n$ of opposite parity with $(m-3)^2 + (n-5)^2 \leq 1$.

**Approved solution**

**(a)** Let $n = 8$. Then $50 - 64 = -14 < 0$. ■
**(b)** Let $x = 5$. Then $25 - 15 - 10 = 0$. ■ ($x = -2$ also works.)

**(c)** Let $m = 3$ and $n = 4$, which have opposite parity. Then $(3-3)^2 + (4-5)^2 = 0 + 1 = 1 \leq 1$. ■

**In (c), check both conditions.** The witness must satisfy the inequality *and* have opposite parity. $m = 3, n = 5$ satisfies the inequality but has the same parity, so it fails.

## Section B — Disproving existence

**2.** Disprove each.

(a) There exists a real number $x$ such that $x^2 + 6x + 10 = 0$.

(b) There exists an integer $n$ such that $n^2 = 3$.

**Approved solution**

**(a)** Let $x \in \mathbb{R}$. Completing the square,

$$x^2 + 6x + 10 = (x^2 + 6x + 9) + 1 = (x+3)^2 + 1 \geq 1 > 0.$$

So $x^2 + 6x + 10 \neq 0$ for every real $x$. ■
**(b)** Let $n \in \mathbb{Z}$. If $|n| \leq 1$ then $n^2 \leq 1 < 3$; if $|n| \geq 2$ then $n^2 \geq 4 > 3$. So $n^2 \neq 3$ for every integer. ■

**Both are universal arguments.** Part (b) uses cases to cover every integer — testing $n = 1$ and $n = 2$ is suggestive but proves nothing about $n = 1000$.

## Section C — Mixed quantifiers

**3.** Prove or disprove each.

(a) For every integer $n$, there exists an integer $m$ with $n + m = 0$.

(b) There exists an integer $m$ such that for every integer $n$, $n + m = n$.

(c) There exists an integer $m$ such that for every integer $n$, $n + m = 0$.

**Approved solution**

**(a) True.** Given $n$, take $m = -n$. The witness depends on $n$, which is allowed.
**(b) True.** Take $m = 0$. Then $n + 0 = n$ for every $n$.
**(c) False.** Suppose such an $m$ existed. Taking $n = 0$ gives $m = 0$; taking $n = 1$ gives $1 + m = 0$, so $m = -1$. Both cannot hold. So no single $m$ works.

**(a) and (c) differ only in quantifier order**, and the order decides the answer. In (a) the witness may depend on $n$; in (c) it must be chosen first. Week 4 again, now as a proof technique.

## Section D — Non-constructive existence

**4.** Prove: there exist irrational numbers $a$ and $b$ such that $a + b$ is rational.

**Approved solution**

Let $a = \sqrt{2}$ and $b = -\sqrt{2}$. Both are irrational (the negative of an irrational is irrational). Then $a + b = 0$, which is rational. ■

**This one is constructive** — we named both numbers. Compare Result 3.38, where the analogous statement for $a^b$ needed a case split and never identified the pair.
When a direct witness exists, use it. The non-constructive route is for when you cannot find one.

**5.** Prove that among any 13 integers, some two leave the same remainder on division by 12.

**Approved solution**

Division by 12 leaves one of 12 remainders: $0, 1, \ldots, 11$. Assign each of the 13 integers to its remainder. With 13 integers and only 12 remainders, some remainder receives at least two integers. Those two leave the same remainder. ■

**This is non-constructive.** The proof guarantees two such integers exist but does not say which — that depends on the integers you were given.
It is also the **pigeonhole principle**, which Result (5.3) previewed in Week 7 and Chapter 8 will name formally.

## Section E — Writing

**6.** In five to eight complete sentences, explain the difference between a constructive and a non-constructive existence proof. Use Result 3.38 as your example of the second, and say why the distinction matters to a computer scientist.

**Approved solution**

**Model answer.** A constructive existence proof establishes $\exists x,\ R(x)$ by producing a specific object and verifying it has the property, as in "let $n = 1$, and then $2 - 1 > 0$." A non-constructive proof shows the object must exist without identifying it. Result 3.38 is the standard example: it considers $\sqrt{2}^{\sqrt{2}}$ and splits into two cases depending on whether that number is rational, obtaining a valid pair in each case. But the proof never determines which case holds, so it tells us that one of two candidate pairs works without telling us which one. For a computer scientist the distinction is substantial, because a constructive proof is essentially an algorithm for finding the object, while a non-constructive proof gives no procedure at all. A program can compute a witness only when some constructive method exists.

**Full credit requires** Result 3.38 explained correctly — including that the proof does not decide the case — and the algorithmic point.

**Instructor notes**

- **Problem 2(b)** shows that disproving existence needs a universal argument even when the answer is obvious. Watch for "I tried 1 and 2."
- **Problem 3** pairs (a) and (c), which differ only in quantifier order. Board time.
- **Problem 5** is the pigeonhole principle appearing early and non-constructively. Worth flagging as a preview of Week 12.
- **Problem 4** contrasts a constructive proof with Result 3.38's non-constructive one. Students should prefer the direct witness whenever one exists.
- Reserved for Homework 10: Section 3.6 even-numbered exercises.